



(Un)usual advocacy coalitions in a multi-system setting: the case of hydrogen in Germany

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Abstract

Grand sustainability challenges span multiple sectors and fields of policymaking. Novel technologies that respond to these challenges may trigger the emergence of new policy subsystems at the intersection of established sectors. We develop a framework that addresses the complexities of ‘multi-system settings.’ Empirically, we explore belief and coalition formation in the nascent policy subsystem around hydrogen technologies in Germany, which emerges at the intersection of electricity, transport, heating, and industry and is characterised by a broad range of actors from different sectoral backgrounds. We find two coalitions: a rather unusual coalition of actors from industry, NGOs, and research institutes as well as an expectable coalition of gas and heat sector actors. Actors disagree over production, application, and import standards for hydrogen. However, there is widespread support for hydrogen and for a strong role of the state across almost all actors. We explain our findings by combining insights from the advocacy coalition framework and politics of transitions: Belief and coalition formation in a nascent subsystem are influenced by sectoral backgrounds of actors, technology characteristics, as well as trust and former contacts. Our study contributes to a better understanding of early stages of coalition formation in a multi-system setting.

Keywords Multi-system interaction · Advocacy coalition framework · Nascent policy subsystem · Politics of transitions · Hydrogen · Energy transition

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Introduction

Grand sustainability challenges such as climate change, loss of biodiversity, or resource depletion pose massive threats to ecosystems and societal well-being. At the same time, many sustainability challenges are particularly difficult for policy to address: they span multiple subsystems of policy-making (Biermann et al., 2012; Brandenberger et al., 2022; Milhorange et al., 2021; Reber et al., 2022), they require policy support (e.g., to stimulate innovation) while being confronted with strong lock-ins around established socio-technical systems (Unruh, 2000), and there are different groups of actors with conflicting views on potential solutions (Avelino et al., 2016; Markard & Rosenbloom, 2020). Grand sustainability challenges do not only confront established approaches in policy analysis (Levin et al., 2012) but also call for new ways of theorising to cope with the underlying complexity (Head & Alford, 2015).

Our study seeks to tackle some of these new complexities. We build on earlier research on the advocacy coalition framework (ACF) and the interdependence of developments in politics and technology (Schmidt & Sewerin, 2017). Markard et al. (2016) suggest that policy subsystems and socio-technical systems influence each other and may co-evolve over time. Drawing on policy feedback theory, Schmid et al. (2019) argue that changes in advocacy coalitions can be explained by feedback from earlier policy outcomes. Here, we take these ideas a step further. We include technology as well as the relevance of ‘sectoral backgrounds’ (e.g., established practices, norms, business models or infrastructures within a sector) to explain the formation of beliefs and coalitions of actors.

Empirically, we focus on hydrogen in the wider context of climate change and the transition to low-carbon energy systems. The energy transition has entered a new stage of development as many countries and regions, including large emitters such as the US, the EU or China, have formulated long-term targets of reaching net-zero greenhouse gas emissions (ECIU, 2021). Net-zero targets are a game changer because they require fundamental socio-technical transformations and policy changes in all sectors. They even necessitate entirely new approaches for decarbonisation such as novel fuels based on hydrogen. As a consequence, the net-zero energy transition confronts policy and politics with new complexities. This multi-system setting (Andersen & Geels, 2023; Löhr & Chlebna, 2023; Ohlendorf et al., 2023; Rosenbloom, 2020) includes (i) transformations across multiple sectors (e.g., electricity, gas, transport, industry), (ii) the emergence of new technologies, and (iii) a large number of actors from different sectoral backgrounds.

Assuming that support by a broad range of actors is crucial for the progress of the net-zero energy transition, we explore how actors position themselves, and form coalitions, when confronted with new, transformative technologies. We argue that, in certain cases, new policy subsystems emerge around new technologies and we build on the advocacy coalition framework (ACF) (Weible et al., 2020) to study them. While recent work has improved our understanding of nascent policy subsystems (Ingold et al., 2017; Stritch, 2015), we still know little about settings characterised by a complex interplay of multiple sectors, technologies and actors. We ask: How do actors form beliefs and coalitions in a new policy subsystem situated in a multi-system setting?

To address this, we develop a conceptual framework that builds on the ACF and the politics of transitions, a field of research to study policy struggles over large-scale system changes (Köhler et al., 2019). The transitions perspective highlights that actors are both

challenged by (technological) innovations and, at the same time, embedded into (and influenced by) broader socio-technical systems, which we refer to as sectoral backgrounds. The ACF analyses changes in beliefs and coalitions to understand policy change within a policy subsystem (Weible et al., 2020). Next to beliefs, the ACF also emphasises the relevance of former contacts when actors enter a nascent subsystem (Ingold et al., 2017; Stritch, 2015). Combining both perspectives, we explain coalition formation both by specific sector and technology characteristics as well as former alliances, trust, and contacts.

Our empirical analysis focuses on novel technologies that emerge around hydrogen. Hydrogen is a relevant case because it can be used to decarbonise many different sectors, including heating, transport, electricity, and hard-to-decarbonise industries such as steel or cement. We study Germany as a country with ambitious policy goals for the development of hydrogen (BMW, 2020) and an important location for several energy-intensive industries.

Methodologically, we apply a mixed methods approach to identify beliefs. We analyse 42 position papers of key actors referring to the German hydrogen strategy, code the documents with QDA software and analyse the data through qualitative content analysis. To identify different advocacy coalitions, we apply a cluster analysis and visualise the results through multidimensional scaling. We also take a closer look at the three most conflictive policy core beliefs. We complement this analysis with insights from nine expert interviews.

We find two emerging coalitions: an expectable alliance of incumbent actors from the gas and heat sector and an unusual coalition of some industry actors, NGOs, and research institutes. While actors of both coalitions share a widespread support about using hydrogen to reach net-zero targets, conflicts emerge around its production, import, and application.

The paper is structured as follows: First, we outline the “[Theoretical background](#)”. In the section on “[Methods and case](#)” we explain our methods including case selection, introduction to our empirical case, data collection, coding and approach combining cluster- and qualitative content analysis. In the “[Results](#)” section, we present our results, including emerging advocacy coalitions and political conflicts. We discuss our findings in our “[Discussion](#)” section and end with a “[Conclusion](#)”.

Theoretical background

Policy analysis has started to address the increasing complexity of policymaking for grand sustainability challenges. Towards this end, scholars have explored different strands of literature within and beyond political sciences to develop new conceptual approaches or enhance existing ones (Blake et al., 2020; Brandenberger et al., 2022; Goyal & Howlett, 2020; Kammerer & Ingold, 2021; Kern & Rogge, 2018; Roberts & Geels, 2019). We do the same. Our study departs from the *advocacy coalition framework* (Sabatier, 1998) and draws on complementary insights from the *politics of transitions* (Köhler et al., 2019). Our goal is to explain the formation of beliefs and new coalitions of actors in a nascent policy subsystem. Our key argument is that, beliefs are not only shaped by prior experiences and contacts (as the ACF suggests), but also by technology characteristics and the sectoral background of the actors (highlighted by the politics of transitions). Next, we briefly introduce and compare both approaches and present our combined conceptual framework.

The advocacy coalition framework

The advocacy coalition framework is a well-established policy process theory to explain policy change (and stability) through the formation and change of beliefs and coalitions of actors (Sabatier, 1998; Weible et al., 2020). According to the ACF, *advocacy coalitions* form based on shared *policy core beliefs* and coordination within a *policy subsystem* (Sabatier & Jenkins-Smith, 1993). Policy core beliefs are subsystem-specific, normative and empirical (Weible & Ingold, 2018). They may change over time (e.g., several years) and lead to major policy change.¹ Different coalitions compete over policy issues as they seek to influence policy. According to the ACF, policy core beliefs of a dominant coalition are eventually translated into policies (in the sense of policy strategies, or policy goals). Below, we revisit key concepts and knowledge gaps related to the particularities of our case: nascent subsystems, multiple subsystems, and the role of technology.

Nascent policy subsystems are emerging subsystems with first policy outputs and actors starting to professionalise in the respective area (Ingold et al., 2017). However, coordination between actors, secondary aspects, and subsystem boundaries may still be in flux (Ingold et al., 2017; Weible et al., 2020), and policy issues may be discussed in different venues (Lemke et al., 2023). Of particular interest in such an early stage of development is the formation of policy core beliefs. Prior research suggests that policy core beliefs are not fully developed and unstable (Nohrstedt & Weible, 2010; Weible et al., 2020). Actors may rely on experts or prior experiences from other policy fields when forming their beliefs (Ingold et al., 2017; Kammerer & Ingold, 2021). For example, factors such as trust or former contacts have been observed to affect coalition formation in nascent subsystems (Ingold et al., 2017). Also, the social influence, e.g., the relationships between actors, can influence belief changes and policy learning in nascent subsystems (Gronow et al., 2021). Overall, coordination between actors might be less established than in mature subsystems (Weible et al., 2020).

A second particularity is that we are confronted with actors from *multiple systems*. In policy research, scholars have started to address cross-cutting or ‘trans-subsystem’ policy challenges such as climate change or migration, widening the focus from single subsystems to interactions among several subsystems (Jones & Jenkins-Smith, 2009; Meckling & Goedeking, 2023; Milhorange et al., 2021; Reber et al., 2022). The literature on policy integration discusses multiple subsystems and sectors (Candel & Biesbroek, 2016; Kefeli et al., 2023; Tosun & Lang, 2017; Trein et al., 2023). Policy research has studied multiple issues, multiple-policy subsystems, an integration of instruments from multiple policy sectors, or actors engaging in multiple debates (Brandenberger et al., 2022; Hedlund et al., 2021; Leifeld et al., 2022). Despite this recent progress, we still lack studies on how actors from different sectoral backgrounds interact as they form coalitions.

A third particularity is about subsystems, in which *technology* and *technological change* play a key role. There are two issues here. First, technology may reach across different subsystems or jurisdictions, which is why it has been termed an “ill-fitting problem” by Lewallen (2022). Second, there is a bi-directional relationship between technology change and policy change (Schmidt & Sewerin, 2017). In relation to the ACF, several studies have shown how technological change (related to renewable energies and low-carbon fuels) has

¹ In addition to policy core beliefs, deep core beliefs are also part of the three-tiered, hierarchical belief system. These fundamental normative convictions are very hard to change (Weible & Ingold, 2018). The very specific, often instrument-related and technical secondary aspects are relatively easy to change and span only limited parts of the subsystem (Sabatier 1998; Weible et al., 2020).

affected beliefs and coalitions (Lindberg & Kammermann, 2021; Markard et al., 2016; Schmid et al., 2019), with some coalitions even centring around specific technologies (Jacobsson & Lauber, 2006; Malmberg, 2023). To be clear, there are different types of policy issues and subsystems: Those in which technological change plays less of a role (e.g. smoking or pensions, Leifeld, 2013; Sato, 1999) and those where it is important. As we deal with a subsystem associated with policy issues around novel technologies, we need to consider (changing) technology characteristics to affect coalition formation. The nature of the policy subsystem also determines which actors participate in the policy process. In ‘technology-heavy’ subsystems, we might see firms and industry associations as central actors seeking to shape policymaking.

Politics of transitions

Transition studies is a growing field of research concerned with the transformation of socio-technical systems (Köhler et al., 2019). We explain it in some more detail for those not familiar with it. Within this field, research on the politics of transitions explores the political struggles over the direction and pace of transformation processes and the associated policy changes (Meadowcroft, 2011; Roberts et al., 2018; Schmid et al., 2021). Actors target policy processes in order to shape the course of a transition in their favour (Lauber & Jacobsson, 2016; Löhr & Mattes, 2022; Markard et al., 2021; Rosenbloom et al., 2018) and they also form alliances or coalitions to leverage their influence on policymaking (Culhane et al., 2021; Hess, 2019a, 2019b). To study the politics of transitions, scholars have combined key concepts from transition studies such as the multi-level perspective (Geels, 2002, 2019) with policy process frameworks (Kern & Rogge, 2018) such as the ACF (Markard et al., 2016) or multiple streams (Derwort et al., 2021). They drew on historical institutionalism (Roberts & Geels, 2019), institutional and transition work (Bjerkan et al., 2021; Löhr & Chlebna, 2023; Löhr et al., 2022; van Doren et al., 2020).

We briefly discuss key concepts and knowledge gaps related to our particular setting: socio-technical systems, multiple systems, (technological) innovation, and sectoral background.

Socio-technical transitions are understood as fundamental and long-term transformations of socio-technical systems (Geels, 2002; Köhler et al., 2019). Scholars also use the term “sustainability transitions,” if the focus is on transformations toward more sustainable modes of production and consumption (Markard et al., 2012). *Socio-technical systems* are semi-coherent sets of actors (e.g., firms, associations, NGOs, policymakers), institutions (e.g., policies, societal norms) and technologies that, together, provide essential services for society (Rip & Kemp, 1998). Socio-technical systems can be defined at various levels of aggregation but in many studies, they are equated with sectors such as energy supply, transport or food that fulfil key societal functions (for a comparison with policy subsystems see 2.3). Below, we primarily use the term sector(s) instead of socio-technical system(s) to avoid confusion with policy (sub-)systems.

Past research has mostly focused on the transition of single sectors (Geels, 2005, 2006) but contexts, in which *multiple socio-technical systems* transform in parallel are receiving increasing attention (Andersen & Geels, 2023; Andersen & Markard, 2020; Löhr & Chlebna, 2023; Rosenbloom, 2020; Schot & Kanger, 2018). The net-zero energy transition is an example of such a multi-system setting: Multiple transitions unfold at the same time in multiple sectors (energy, transport, buildings, industry) and the success of the overarching transition depends on the various individual sectoral transformations

complementing each other, all the while adverse developments are equally conceivable (Markard & Rosenbloom, 2022; Ohlendorf et al., 2023).

Innovations are central to transitions (Elzen & Wiecek, 2005; Smith et al., 2010). Together with exogenous developments (e.g., shocks), they are a key driver of system transformations. In the transitions literature, new technologies are often in the focus of empirical studies, but also non-technical innovations (e.g., institutional changes, new types of organisations, new practices) are important. Innovations emerge in niches (Geels, 2002) and, once they have matured, they may diffuse widely, thereby leading to the decline of existing technologies and practices (Koretsky et al., 2022). Transitions involve focal innovations (e.g., electric vehicles) but also complementary innovations (e.g., charging stations, batteries) (Andersen & Markard, 2020). In the case of multi-purpose technologies such as those around the energy carrier hydrogen, complementary innovations vary across sectors (Ohlendorf et al., 2023). For our study, it is important to note that the characteristics of both focal and complementary innovations influence the policy positions of actors. Policy positions may change, for example, as technologies mature (Markard et al., 2016).

Another key lesson from the politics of transitions is that actors are embedded into existing socio-technical systems (Geels, 2020). This *sectoral embedding* (or *sectoral background*) means that their activities and positions are shaped, i.e., facilitated and constrained, by established practices, norms, infrastructures, technologies etc. At the same time, they are challenged but also attracted by innovations. So, they are confronted with a tension between existing structures and potentially new ones. For our study, we need to consider that actors come from different sectors, each characterised by different practices, technologies, and transition challenges (Ohlendorf et al., 2023). We expect that these sectoral backgrounds shape the policy positions of actors, and—eventually—the coalitions they build.

In terms of actors, transition research distinguishes between incumbent actors (e.g., established businesses) which often see transitions as threats (van Mossel et al., 2018) and niche actors or challengers (e.g. start-ups) which seek to foster transitions. Incumbents have been observed slowing down ongoing transitions (Markard & Rosenbloom, 2020; Smink et al., 2015) but also driving innovation (Berggren et al., 2015; Kishna et al., 2017).

Contrasting the advocacy coalitions framework and politics of transitions

Here, we juxtapose the ACF and politics of transitions (Table 1) to make similarities and differences explicit and prepare our combined conceptual approach. Key terms and their definitions may differ; we link those that are comparable.

General approach (policy and technological change): Both frameworks are concerned with how actors affect policy processes and how eventually, they might bring about policy² change. In the ACF, changes in policy core beliefs indicate a major policy change, changes in secondary aspects a minor policy change. Both approaches theorise (system) stability, i.e., why change does not occur, as well as the conditions under which it does. In the politics of transitions, policy changes unfold in the context of larger transition processes, in which technological change is also key. The ACF, in contrast, can be applied to a broad range of policy issues, whether related to transitions and technology change or not. The ACF takes a mid-term perspective (e.g., a decade, often addressing one case of policy change) while

² In both frameworks, policies are understood as the substance of policymaking, including general objectives, programmes, strategies, regulations, or funding priorities.

Table 1 Comparison of the advocacy coalition framework and politics of transitions

	Advocacy Coalition Framework	Politics of Transitions
<i>General approach</i>	Explain major (or minor) policy change or stability; can be applied to a broad range of policy issues; technological change is not key	Explain political struggles and socio-technical change including policy change in the context of larger socio-technical transitions; technological change is key.
<i>System emergence</i>	Subsystems emerge due to external shocks, new issues on the political agenda, dissatisfied actors	Systems change as a result of external shocks, systems emerge due to innovation; over time, systems mature and decline (lifecycle)
<i>System concept</i>	Policy subsystem: Set of actors engaging with the same policy issue, influencing policy subsystem affairs Subsystem is largely stable	Socio-technical system: Actors, institutional structures and technologies that together fulfil a societal function System is largely stable
<i>Policy</i>	General policy strategies, measures or regulations that are influenced by policy core beliefs	General policy strategies, regulations and policy instruments that are influenced by policy positions
<i>Policy positions and beliefs of actors</i>	Policy positions of actors are shaped by the three-tiered belief system including deep core beliefs, policy core beliefs and secondary aspects; experts and prior experiences matter in nascent subsystems	Policy positions are shaped by various factors, including established practices and prior experiences but also opportunities / threats arising from innovations
<i>Coalitions</i>	Actors form coalitions based on shared policy core beliefs and coordination	Originally no explicit conceptualisation of coalitions, typically dichotomy between incumbents and challengers

transition perspectives have a long-term orientation (several decades, which might cover several major policy changes), even though studies often look at shorter periods. While technological change is key in politics of transitions, it is less central in the ACF.

System concept Both frameworks apply a system concept but there are differences. The policy subsystem is defined around a specific policy issue and a geographical scope, often a country (Sabatier & Jenkins-Smith, 1993; Weible & Ingold, 2018). The key elements are individual and collective actors forming coalitions. Typical ACF actors are policymakers, governmental officials, interest organisations or scientists that engage in the policy subsystem affairs. The socio-technical system includes actors, institutional structures, and technologies as key elements, which together fulfil a societal function (e.g., provision of transport). Typical ‘transition actors’ are firms, associations, NGOs and policymakers. Like the policy subsystem, the socio-technical system has a topical focus (typically defined as a sector) and a spatial boundary (often national level) (Köhler et al., 2019; Markard et al., 2012). Policy subsystems are considered to be stable over time, although this view is challenged in the analysis of nascent subsystems which are not stable yet. The emergence of a new subsystem may cause changes in adjacent policy subsystems. Socio-technical systems are rather stable as well, unless disrupted by a transition.

System emergence In the ACF, a policy subsystem emerges due to external shocks, new issues on the political agenda or actors being dissatisfied with the way an issue is treated in an existing subsystem (Beverwijk et al., 2008; Weible & Ingold, 2018). Socio-technical systems change as a result of external shocks in combination with innovation (Geels, 2002;

Geels & Schot, 2007). System emergence is part of an entire lifecycle: innovations develop in niches, mature, diffuse widely and eventually decline (Markard, 2020).

Policy Both approaches address politics and policy. The ACF posits that the policy core beliefs of a dominant coalition are later translated into policies (Sabatier, 1998; Weible & Ingold, 2018). Policies include general strategies and measures, regulations, or policy instruments that address policy problems. In the ACF, policy outcomes result from the interplay of advocacy coalitions competing over different problem definitions (in the form of beliefs) that have undergone formal governmental decision-making. In the politics of transitions, politics in the form of struggles over vested interests and contested views on system change has been at the heart of theorising and analyses (Kern & Rogge, 2018). These power struggles, often between incumbents and challengers, influence policy outputs and may eventually lead to policy changes. While policy (or policy change) may rather be analysed as the dependent variable in the ACF, the politics of transitions see changes in policy, technology and organizations as co-evolutionary.

Policy positions of actors In the ACF, policy positions are modelled through the concept of beliefs: secondary aspects, policy core beliefs, and deep core beliefs (Weible & Ingold, 2018). The more specific a position is, the more likely the actor is to change it (e.g. secondary aspects). Policy core beliefs encompass positions on general policy solutions and policy instruments which can also be technology-related (Weible & Ingold, 2018), e.g. a technology such as hydrogen is seen as a decarbonisation solution. Actors engage in policymaking to translate these beliefs into policies. In transition studies, policy positions are shaped by a variety of factors, including the sectoral embedding of actors: existing norms, practices, business models, or infrastructures (see above). Actors might be torn between existing assets and new (transition) challenges, including the use of new technologies.

Coalitions In the ACF, beliefs (and coordination) are key to identify advocacy coalitions as well as conflict and consensus about policy issues between coalitions (Sabatier, 1998). Recent ACF literature further differentiates coalition ideal types such as disconnected, ephemeral or adversarial advocacy coalitions. In case of the latter “policy actors share beliefs, coordinate their activities (...), use their resources competitively, and do all of this over extended periods of time” (Weible et al., 2020, p. 1067). Several empirical applications found coalitions for which technology-related beliefs matter (Jacobsson & Lauber, 2006; Lindberg & Kammermann, 2021; Malmborg, 2023; Schmid et al., 2019). The ACF mainly sees policy actors (e.g. interest groups, government officials or scientists) engaging in coalitions and influencing decision making (Weible et al., 2020). Actors may take different roles such as the policy broker who negotiates a compromise between coalitions. The politics of transitions start with a diverse set of actors engaging in politics, covering the policy as well as technological and industry dimensions. Roles of actors in coalitions are—apart from the challenger incumbent dichotomy—less clearly defined. Nevertheless, in the politics of transitions, the role of coalitions is increasingly acknowledged (Roberts et al., 2018), e.g. through applications of the ACF or discourse coalitions (Haukkala, 2018; Hess, 2019a; Lowes et al., 2020; Markard et al., 2016) but not as explicitly conceptualised as in the ACF.

Our key motivation to combine both perspectives is that they can complement each other in explaining belief and coalition formation in a nascent policy subsystem which emerges at the intersection of several sectors. Actors may cooperate for the first time, originate from different backgrounds (here: adjacent sectors) and bring different experiences. The transition perspective offers socio-technical explanations (concerning

sectors and technologies) for how beliefs and coalitions form, while the ACF highlights trust and former contacts (Ingold et al., 2017).

At the intersection of the advocacy coalition framework and politics of transitions

The core element of our conceptual framework is a nascent policy subsystem³ emerging at the intersection of established (but transforming) socio-technical systems, or sectors (Fig. 1). In the subsystem, actors debate new policy issues that arise as one or more transitions unfold.

In the new policy subsystem, we expect to find many actors that were (and still are) active in other policy subsystems⁴ and/or in the adjacent sectors. These are depicted in blue colours. There will also be new actors (e.g., new organisations), depicted in white. We expect those *actors* to join that have a stake in the novel technology and seek to shape future policies. Firms and industry associations from established sectors may see new business opportunities but they might also be challenged by the innovation. Policymakers, NGOs or think tanks may enter as well if they see potential in the novel field or have concerns.

We expect that actor beliefs and the formation of coalitions are shaped by several factors. These include the characteristics of the *novel technology* (white triangle),

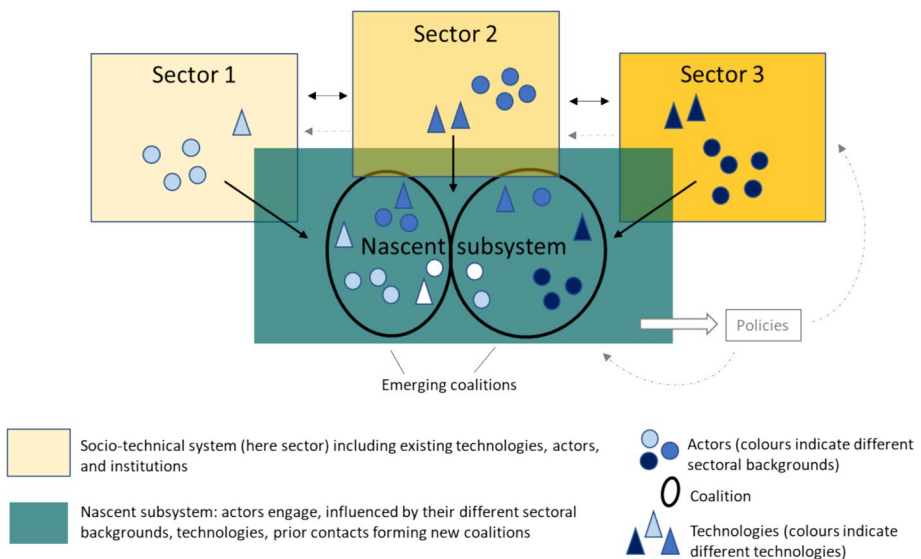


Fig. 1 Conceptual framework linking key elements of politics of transitions and the ACF to explain belief and coalition formation.

Actors forming coalitions in the nascent subsystem are influenced by different sectoral backgrounds and technologies which determine the newly emerging beliefs and eventually policies in the nascent subsystem. The figure shows a snapshot of a subsystem that may change over time

³ We use the subsystem notion as the established term in the ACF. This does not imply that the policy subsystem is a subsystem of the socio-technical systems, or sectors.

⁴ For the sake of simplicity, these are not depicted in our figure.

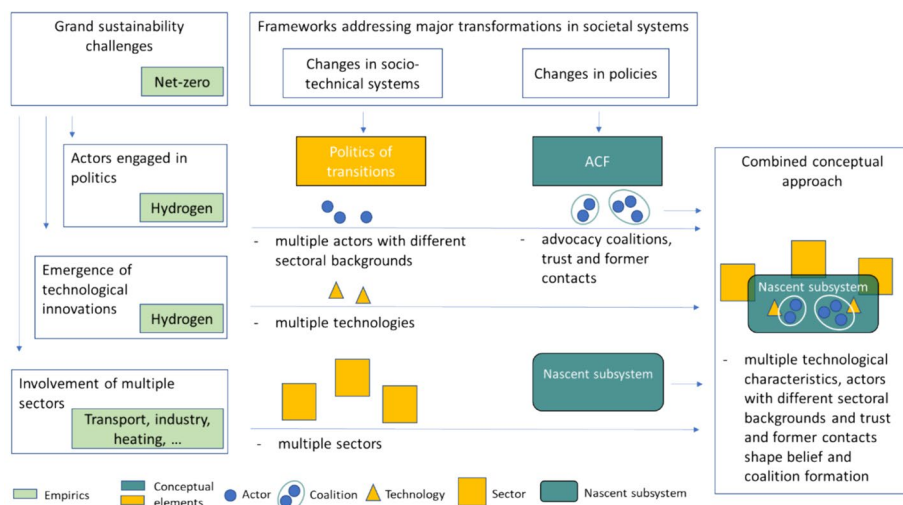


Fig. 2 Conceptual framework on belief and coalition formation in a nascent policy subsystem in a multi-system setting.

This figure assembles the conceptual elements we use from both perspectives. It highlights multiple technologies and sectors as key elements from the politics of transitions (in yellow) and coalitions, trust, former contacts and nascent subsystems from the ACF (darker green). Actors (in blue) appear in both frameworks (color figure online)

the characteristics of existing sectoral technologies (blue triangles) and the sectoral background of actors (circles in different shades of blue). We also expect trust and former contacts (not visualised in the figure) to affect coalition formation.

Figure 2 is a conceptual figure with empirical elements. It illustrates how we have built our framework (right column) based on the inputs of the two perspectives (columns in the middle, with yellow and green colours) and the particularities of our empirical context (left column, different levels). The net-zero energy transition is characterised by a complex interplay of multiple actors with different sectoral backgrounds, multiple technologies, and multiple sectors.

In this context, nascent subsystems emerge, in which actors form coalitions. Belief and coalition formation are shaped by (i) trust and former contacts (from the ACF, no special symbols), (ii) technology characteristics (here related to hydrogen), and (iii) the sectoral backgrounds of the actors. Through the combination of these factors, our framework can not only explain ‘usual coalitions’ (e.g., of actors that have already collaborated in the past) but also ‘unusual coalitions,’ where actors from different sectors form a coalition because they have very similar views on the novel technology.

Methods and case

This chapter describes the selection of our case, our data, and methods. We follow a mixed-method approach based on a qualitative content analysis of position papers, complementary semi-structured expert interviews and a cluster analysis.

Case selection: hydrogen in Germany

Given our interest to study a nascent policy subsystem in a multi-system, net-zero context, we decided to focus on hydrogen: a set of novel technologies and political activities that are potentially relevant to decarbonise different sectors, including hard-to-decarbonise industries (e.g., chemicals, steel, cement), electricity, mobility, and heating. Hydrogen is in an early stage of development and actors from the above sectors show a strong interest, start to position themselves and ask for political support.

Next, we decided to study Germany as a country with net-zero targets (Deutscher Bundestag, 2021), ambitious policy goals for the development of hydrogen (BMW, 2020), and an important location for several energy-intensive industries. In Germany, the net-zero energy transition is in an acceleration phase, which is characterised by a decline in established nuclear and fossil fuel technologies, an increase in renewable energy technologies and stronger system integration (Löhr & Mattes, 2022; Markard, 2018). These changes go along with many challenges and frictions and on all levels, such as contestation around further renewable energy expansion, energy prices, or the speed of coal phase-out (Chlebna & Mattes, 2020; Löhr et al., 2022; Rogge & Johnstone, 2017).

In this situation, hydrogen comes in as a supposedly attractive, complementary innovation as it is a storable and transportable energy carrier and climate neutral if generated from renewable energies.⁵ Recently, hydrogen gained particular attention—politically and in industry—as potential solution to address hard-to-decarbonise industries such as aviation, shipping, chemicals, cement or steel (Belova et al., 2023; Kern et al., 2023).

We pursue a single case study on hydrogen in Germany. This allows for an in-depth analysis of actors involved in the process, their policy positions and beliefs. Due to the early stage of development, a case study allows us to answer ‘how’ questions (Yin, 2018): how beliefs and coalitions form in a new subsystem.

Hydrogen as a nascent policy subsystem

A policy subsystem is determined by a substantive focus (here: new policies for hydrogen), a geographical scope (here: Germany and its federal government, which has the ability to implement ambitious hydrogen policies) and a set of actors (see below) that engage in subsystem affairs (Sabatier & Jenkins-Smith, 1993; Weible & Ingold, 2018). To identify a nascent policy subsystem, Ingold et al. (2017) suggest three criteria: (1) a lack of final policy outputs, (2) first actors specialising in the new policy area and (3) a great belief diversity concerning policy core beliefs and secondary aspects. Here, we discuss the first two points. The third will be assessed in the results section.

First, in the past, hydrogen was associated with fuel cell applications in transport and heating. In Germany, we find early policy outputs such as an R&D programme from 2007. However, a comprehensive hydrogen policy strategy targeting multi-sectoral uses has not been formulated until recently. In 2019, a new dynamic started to unfold with first hydrogen strategies being published by the federal states (Flath et al., 2023), numerous events, including a national hydrogen conference organised by four different federal ministries

⁵ Mostly mentioned in the debate about hydrogen are the following three ‘hydrogen colors’: ‘Green’ hydrogen is produced through electrolysis by renewable sources, ‘blue’ hydrogen is produced from steam reduction of natural gas combined with carbon capture and storage (CCS). ‘Grey’ hydrogen is generated from steam reforming of fossil fuels and the CO₂ is released into the atmosphere.

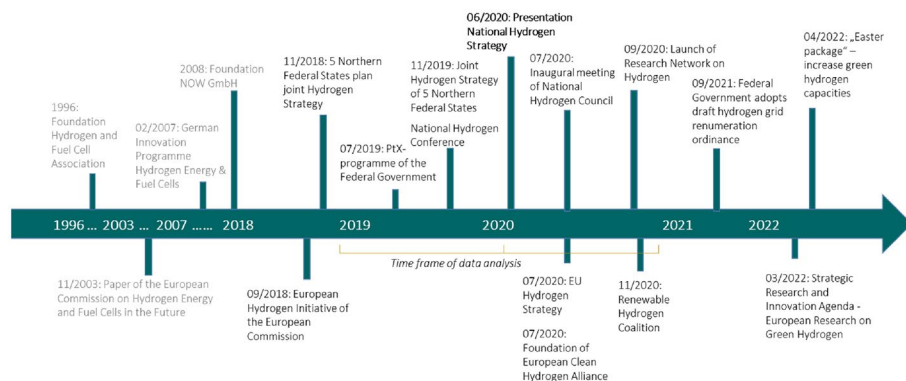


Fig. 3 Key policy events and creation of associations targeting the development of hydrogen in Germany and the EU (color figure online)

(Fig. 3). The involvement of different ministries points to the multi-system character of hydrogen, i.e. existing jurisdictional boundaries do not fit the new policy issues (Lewallen, 2022). The hydrogen conference was key for the political process toward the German hydrogen strategy, which was adopted in June 2020 and qualifies as first major policy output. With its hydrogen strategy, the German government seeks to establish a (hardly existing) hydrogen market (supported with 7 billion euros) and build international partnerships for hydrogen imports (another 2 billion euros). Until 2030, 5GW of hydrogen production capacity shall be installed, another 5GW until 2035.⁶ The focus of the German government is on green hydrogen, while blue hydrogen is seen as a bridge technology.

The hydrogen strategy is the focal event for our paper.⁷ Our analysis covers the year preceding its publication and position papers published until December 2020. Even after the publication of the strategy, many policy issues such as application in different sectors, grid use, international agreements, import standards etc. still need to be clarified. Many of these decisions are likely to be contested, which is why our study offers important insights. While the hydrogen strategy qualifies as a first policy output, there is no hydrogen law to date, which shows that the policy subsystem is only just emerging.

Second, shortly before and around the publication of the hydrogen strategy, more and more actors began to engage with hydrogen, e.g., creating networks like the research network on hydrogen or the European Renewable Hydrogen Coalition. Also, the German hydrogen and fuel association (DWV) increased its activity, e.g. publishing three position papers since 2019 in contrast to only one before 2019. Along with the publication of the hydrogen strategy actors that did not have much expertise on hydrogen started to focus on the new area, e.g. NGOs:

⁶ The 2021 coalition agreement foresees 10GW until 2030.

⁷ We do not concentrate on subsequently adopted regulation such as the 2021 amendment to the German energy law (EnWG) (see 3.3). It adds regulation on hydrogen grids and non-discriminatory access. The 2021 amended renewable energy law (EEG) contains regulation on hydrogen, providing under certain conditions exemption from the EEG levy for the production of green hydrogen. The update of the hydrogen strategy, published in July 2023, is not part of our analysis but we link back to it in our discussion.

Then, there are also the stakeholders, such as all the NGOs and BUND and WWF and Greenpeace and so on, who have an issue with this import topic and of course want to take a closer look at it, but some of whom are also only just starting to get really deep into it, [...] who are [...] starting a bit later than the technology and economic policy steps that have already been taken. (EXP1, Research institute)

While some industry actors already had some expertise on hydrogen, NGOs, research institutes and associations (e.g. on renewable energies) only started to engage with the topic around 2020.

Procedurally, the process [on developing a hydrogen strategy] in Germany has been strongly influenced by the Gas 2030 Dialogue of the [energy ministry]. ... NGOs weren't really planned at first, until at some point [an NGO] said: "Friends, it doesn't work like that. I want to get involved here too." (EXP6, Think tank)

We thus observe a growing number of actors specialising in the topic, new networks and associations formations as well as publications. This indicates that the policy subsystem is still in its formation phase.

Data collection

To identify policy core beliefs, we analysed position papers and press releases of key actors related to the national hydrogen strategy that have been published between October 2019 and December 2020.

To identify the most influential actors subject to our analysis, we followed a three-step approach. First, we prepared a list of around 80 actors working on hydrogen, including companies, (industry) associations, NGOs, and research institutes. Second, following the reputational approach (French, 1969), we conducted nine interviews with hydrogen experts selected by visibility in the debate and recommendation between February and April 2021. We asked each expert to rank the actors on our list on a three-step scale (very important, important, less important) with respect to hydrogen in Germany. We also asked the experts to add missing, but "very important" actors. This led to a list of 63 actors.

In a third step, we excluded all actors without position paper or press releases on the hydrogen strategy and included those actors that were part of joint position papers (but were not on our list before). Our final sample covers 46 actors (with 42 position papers). Most actors are (industry) associations or companies; some published joint position papers (see Table 2).⁸ Our cluster analysis includes 33 position papers that contain sufficient data to identify beliefs (see section "Coding and analysis").

Finally, the interviews also aimed at better understanding hydrogen politics in Germany: We asked about consent and conflicts about the use, production or imports of hydrogen and about emerging coalitions around hydrogen. Six interviews were recorded and transcribed, extensive notes exist for the remaining three (Table 3).

⁸ The joint position papers include 4 actors we did not have in our initial actor list. These are marked in italics: VEA, VIK, SET and Wirtschaftsvereinigung Stahl

Table 2 List of key hydrogen actors in our study

Actor type	No. of position papers	Actors
<i>Companies</i>	11	50Hertz, Amprion, MAN Energy Solutions, RWE, Salzgitter, Siemens Energy, Tennet, Thüga, Thyssenkrupp, Vattenfall, Wintershall
<i>Associations</i>	18	BDEW, BDH, BDI, BEE, BWE, dena, DIHK, DVGW, DWV, FNB Gas, NOW, OGE, PtX Allianz, VCI, VDA, VDI, VDMA, VKU
<i>Research institutes and think tanks</i>	5	DLR, EWI, Fraunhofer IEG, FZ-Jülich, Öko-Institut
<i>NGOs</i>	4	BUND, DUH, Klima Allianz, WWF
<i>Joint position papers^a</i>	4	• Gas actors: Gascade, OGE, Thüga, Uniper, VCI, VIK, Wintershall • DWV, VDMA, VKU • IG BCE & VCI • Industry associations: BDEW, DVGW, FNBGas, SET, Wirtschaftsvereinigung Stahl, VCI, VEA, Zukunft-Erdgas

Table 3 List of interview partners

Code	Actor group	Ranking of actor list	Transcription
EXP1	Research institute	x	x
EXP2	Research institute	x	x
EXP3	NGO	x	x
EXP4	NGO	x	x
EXP5	Research institute		x
EXP6	Think tank		x
EXP7	Research institute	x	
EXP8	NGO	x	
EXP9	Ministry		

Coding and analysis

We developed the coding system deductively based on the ACF (Sabatier, 1998; Weible & Ingold, 2018), where policy core beliefs are defined as subsystem-specific, normative and empirical. They concern the problem severity, positions on general policy solutions and policy instruments (Weible & Ingold, 2018). They may include policy positions (Ingold et al., 2016). We adapted these to hydrogen and derived an initial set of ten policy core beliefs. These were later reduced to eight as we dropped two (on system structure and alternative solutions) for which we could hardly identify relevant mentions. We discussed the beliefs and their respective four subcodes in several rounds of iteration within the author team. Two authors coded these beliefs with QDA software for a subset of five position papers to further refine the initial set of policy core beliefs and subcodes. Table 4 shows the final coding scheme. Each policy core belief includes a lead question and is subdivided into four subcodes, covering the range of positions on the topic. For example, “hydrogen as a solution” addresses the importance of hydrogen as a response to climate change. Subcodes range between “Hydrogen is not important” and “Hydrogen is very important.”

The final coding scheme was applied to all 42 position papers. To improve reliability, all authors co-coded all codes that received only one or diverging codings. We discussed these passages in detail and adjusted the coding where needed.

We use a cluster analysis to identify advocacy coalitions. Each subcode was assigned a number from 1 to 4. The mean value by actor and policy core belief was then used to perform a hierarchical cluster analysis which we performed with Stata. We compared

Table 4 Policy core belief dimensions and codes

	Policy core belief (N)	Lead question	Sub-codes	Mean / Standard Deviation
<i>Problem severity, cause and solution</i>	Severity of the problem (41)	How important is a (political) response to climate change?	CC1: Climate change is one problem among others. CC2: Climate change is somewhat important. CC3: Climate change is important, net zero by 2050. CC4: Climate change is very important, there is an urgent need to decarbonise.	3.10/ 0.41
	Hydrogen as solution (42)	How important is hydrogen as a response to climate change?	H1: H ₂ is not important. It is the last resort after all other options have been exhausted. H2: H ₂ is somewhat important but other pathways than H ₂ preferred. H3: H ₂ is important but one solution among many. H4: H ₂ is very important.	3.15 0.68
<i>Role of government in subsystem affairs</i>	Role of government vs. market (42)	How important is the role of the state (vs. market) when developing hydrogen technology and infrastructure?	SM1: The state is not important. Markets will do it alone. SM2: The state is somewhat important. It is needed to fund the H ₂ economy in transition period. SM3: State is important. It is needed to lay regulatory foundations. SM4: State is very important. It should play a strong role in developing H ₂ .	3.15 0.49
<i>General goals for policy subsystem</i>	Expansion of renewables for hydrogen production (solution) (39)	How important is the (domestic) expansion of renewables to produce hydrogen?	R1: Expansion of renewables is not important. R2: Expansion of renewables is somewhat important. R3: Expansion of renewables is important. R4: Expansion of renewables is very important.	3.25 0.77
<i>Position on general policy solutions and instruments</i>	Production: Green H₂ vs blue (38)	How should hydrogen be produced? (Importance of green H ₂)	GB1: Green H ₂ is not important. Any H ₂ will do. GB2: Green H ₂ is somewhat important. But blue H ₂ is a viable option as well. GB3: Green H ₂ is important. Blue H ₂ only temporarily. GB4: Green H ₂ is very important. It should be the only option to produce H ₂ .	3.01 1.01
	H₂ imports (32)	Under which conditions should hydrogen be imported?	I1: Conditions for imported H ₂ are not important. I2: Conditions are somewhat important. There should be minimum standards. I3: Conditions are important. There should be high standards for imported H ₂ (env, soc, pol). I4: Conditions are very important. There should be high standards and we should avoid H ₂ imports as much as possible.	2.22 0.94
	Gas infrastructure (26)	How important is it to use the existing gas grid?	G1: It is very important to use the existing natural gas grid (and foster it) for H ₂ . G2: It is important. Grid adjustments may be needed. G3: It is somewhat important. The existing gas grid is of limited use. G4: It is not important. Separate or new infrastructure may be needed.	1.86 0.52
	Sectors (38)	How important is it to prioritize which sectors can use hydrogen?	S1: Prioritisation of sectors is not important. H ₂ can be used in all sectors, including heating. S2: Prioritisation of sectors is somewhat important. H ₂ can also be used for individual transport. S3: Prioritisation of sectors is important. H ₂ should be used primarily in hard-to-decarbonise sectors. S4: Prioritisation of sectors is very important. H ₂ can only be used in hard-to-decarbonise sectors.	2.11 1.04

Source: Policy core beliefs based on Weible and Ingold (2018) and adapted to our case. Policy core beliefs on production, imports, gas infrastructure and sectors reflect general goals and positions on policy solutions. N is the number of actors for which results could be obtained. The policy core beliefs with the highest standard deviation are analysed separately as “**Most conflicting policy core beliefs**” in the respective

Table 4 (continued)

section

the sensitivity of the results with respect to similarity and dissimilarity measures, which provide roughly similar outcomes, not affecting our main findings. We furthermore test the effect of missing data points by reducing the number of included policy core beliefs from eight to six, and by imputing missing observations for actors with up to two missing values. Missing values are replaced with the mean value of the respective policy core belief across all other actors. Both approaches increase the number of actors included to the analysis, first by reducing the amount of required but also used information, second via extrapolating information.

Our main specification considers all eight policy core beliefs, allows for up to two imputed values per actor, and thereby covers information from 33 position papers. This specification applies the average linkage method and the Euclidean distance measure. Our main findings (see next section) are robust over the applied similarity measure.⁹ We create and analyse dendrograms to identify clusters of actors (see fig. 6 in the appendix). We select the two largest clusters with the greatest distance to one another. We also discuss one cluster dominated by companies and associations from the gas sector that has additionally been identified in our interviews but does not represent the third largest cluster. The results are visualised via multi-dimensional scaling that illustrates the belief distance between actors in a two-dimensional graphic (see Fig. 4). Multi-dimensional scaling translates the eight policy core beliefs and their corresponding dimensions used in the cluster analysis into two dimensions. We present the findings in the section on “[Emerging advocacy coalitions](#)”.

To better understand which potential conflicts emerge in the nascent subsystem, we analyse particularly conflicting policy core beliefs. We therefore zoom into the three policy core beliefs with the highest standard deviations of actor codes, namely hydrogen production, application of hydrogen in different sectors and imports. We plot the mean belief values of all actors in a two-dimensional graphic with three policy core beliefs as axes (see Fig. 5). This shows in which specific positions actors disagree most. Our conceptual framework contributes to explain why actors build their beliefs. The findings are shown in Sect. 4.2.

Results

In the following, we present the results of our cluster analysis and zoom into three beliefs, which point to emerging conflicts.

Emerging advocacy coalitions

We find two emerging coalitions (see Fig. 4, Tables 5, 6 and the dendrogram in the appendix). The first and smaller coalition, hereafter “green coalition” (left side, green circle),

⁹ We further ran sensitivity checks and tested different cluster algorithms (single linkage which serves to determine outliers and ward linkage) and distance measures (squared euclidean distance and absolute-value distance (Manhattan)).

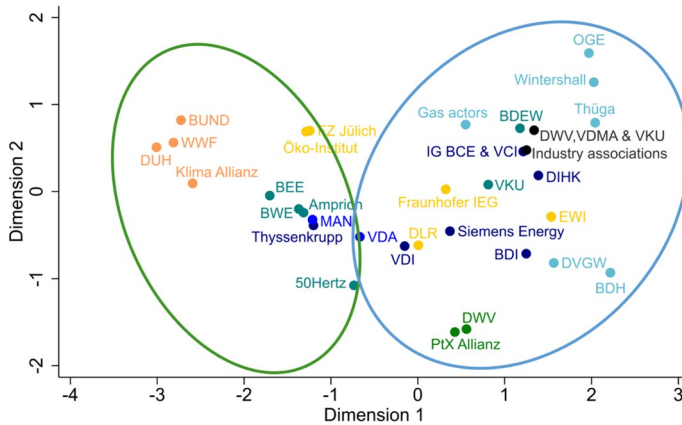


Fig. 4 Emerging hydrogen advocacy coalitions.

Relative distance of actors based on their policy core beliefs. Colours indicate the sector of origin as shown in Table 5. The two dimensions result from multi-dimensional scaling (see section "Coding and analysis") (color figure online)

includes NGOs, renewable energy associations, some research institutes and a few companies. The second coalition is bigger and consists of energy and industry associations, some research institutes, gas sector actors and actors devoted to hydrogen or fuel cells. It is positioned on the right side, hereafter “blue coalition” (blue circle). Belief distance and heterogeneity in the blue coalition are larger than in the green coalition. Consequently, some blue coalition actors, e.g. the automobile association VDA or the Association of German Engineers VDI, are closer to some green coalition actors than to their own coalition allies. We identify a third group of actors as a subset of the large blue coalition. These include the gas transmission grid operator OGE, the ‘gas actors’, the gas company Wintershall, the energy and water association BDEW or the energy and water supplier Thüga (see upper right in Fig. 4 and dendrogram in appendix) (Table 6).

Overall, the distance between the two coalitions is not very large and coalition boundaries are not clear cut. Beliefs are not much polarised but rather distributed heterogeneously, pointing to a greater belief diversity not only between but also within coalitions. This is to be expected in a nascent policy subsystem where beliefs are still forming (see section on “Hydrogen as a nascent policy subsystem”).

Qualitatively, we find that most actors welcome and support the hydrogen strategy of the German government. There is a high degree of consensus that climate change is important and net-zero should be achieved by 2050.¹⁰ Most actors argue that hydrogen is either one solution among many or very important for decarbonisation.

Interestingly, most actors of both coalitions also call for the state to play a strong role in supporting the development of hydrogen technologies and implement the necessary regulatory foundations. While one would expect especially industry actors to opt for market-based solutions (without much state intervention), the early phase of hydrogen development seems to be the reason for the manifold calls for a regulatory basis, whose

¹⁰ This common understanding seems to become a new common sense and thereby diverging from beliefs just a few years ago (Löhr 2020; Leipprand et al., 2017; Markard, Suter and Ingold 2016).

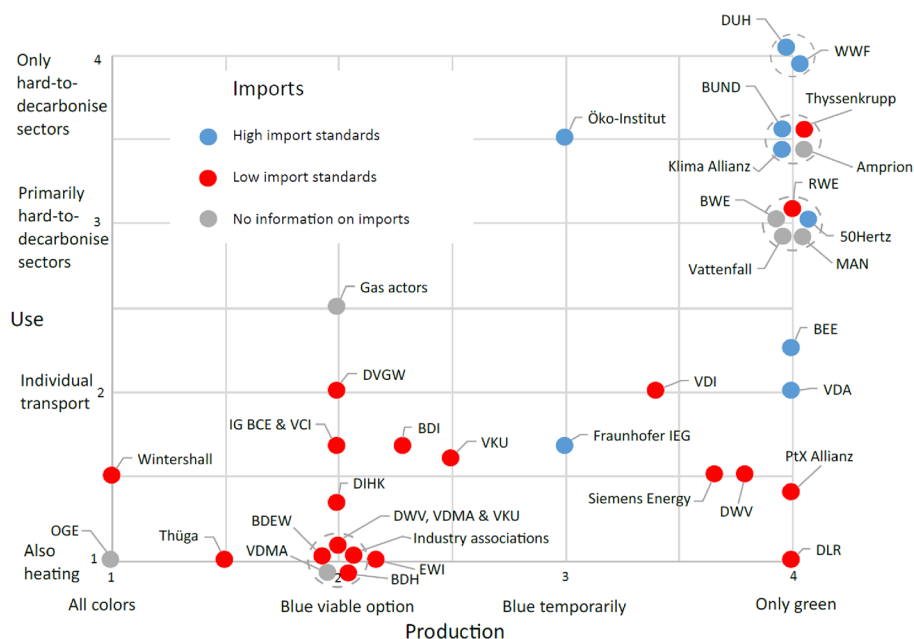


Fig. 5 Policy core beliefs regarding the production, application and imports of hydrogen.

The figure shows the three most conflictive belief dimensions on hydrogen production (x-axis), application (y-axis) and imports (coloured dots). Blue dots were accorded when an actor reached above average codings for import standards (more than 2, 5, meaning high import standards are important) and red dots for below average codings (less than 2, 5 meaning less attention should be paid to import standards). Dotted lines show several actors sharing the same position. Please note that more actors than in the cluster analysis were integrated in this figure as only three positions were needed (color figure online)

Table 5 Actors per sector

Sector	Actor
Industry	BDI, DIHK, Salzgitter, Siemens Energy, Thyssenkrupp, VCI, VDI, VDMA • IG BCE & VCI
Electricity	50Hertz, Amprion, BEE, BWE, RWE, Tennet, Vattenfall
Mobility	MAN Energy Solutions, VDA
Gas / heat	BDH, DVGW, FNB Gas, OGE, Thüga, Wintershall • Gas actors: Gascade, OGE, Thüga, Uniper, VCI, VIK, Wintershall
Hydrogen	DWV, NOW, PtX Allianz
Multiple sectors	BDEW, VKU • DWV, VDMA, VKU • Industry associations: BDEW, DVGW, FNBGas, SET, Wirtschaftsvereinigung Stahl, VCI, VEA, Zukunft-Erdgas
Research / think tanks	Dena, DLR, EWI, FZ-Jülich, Fraunhofer IEG, Öko-Institut
NGOs	BUND, DUH, Klima Allianz, WWF

Please note that some actors pursue activities in more than one sector (e.g. BDEW and VKU are active in the electricity and gas sector)

foundations were only laid after the consultation with the publication of the national hydrogen strategy.

Table 6 Actors and actor types per coalition

Actor type	Green coalition	Blue coalition
Companies	50Hertz, Amprion, Thyssenkrupp, MAN Energy Solutions	Siemens Energy, Thüga, Wintershall
Associations	BEE, BWE	BDI, DIHK, BDH, VKU, BDEW, OGE, PtX Allianz, VDI, VDA, DWV, DVGW
Research institutes and think tanks	Öko-Institut, FZ-Jülich	DLR, EWI, Fraunhofer IEG
NGOs	BUND, DUH, Klima Allianz, WWF	
Joint position papers (associations and companies)		- Gas actors: Gascade, OGE, Thüga, Uniper, VCI, VIK, Wintershall - DWV, VDMA, VKU - IG BCE & VCI - Industry associations: BDEW, DVGW, FNBGas, SET, Wirtschaftsvereinigung Stahl, VCI, VEA, Zukunft-Erdgas

Despite this broad consensus, there are also differences between the two coalitions. We present general differences below and the three emerging conflicts on hydrogen production, application and import in the section on "[Most conflicting policy core beliefs](#)".

Actors in the green coalition give greatest importance to a (political) response to climate change but do not necessarily see hydrogen as the main or only solution to respond to the decarbonisation challenge. Green coalition actors stress the need for energy efficiency and electrification.

(...) the environmental associations (...) are not interested in the volume, but rather think: Oh, better not so much at first or at all. (EXP1, Research institute)

Significantly expanding renewable energies—for direct electrification as well as green hydrogen production—is a core belief of this coalition, which would also rather produce hydrogen domestically, define high standards for imports and rather construct a new hydrogen infrastructure than expanding the existing natural gas grid. This coalition regards hydrogen as an expensive and scarce good, mainly to be used in hard-to-decarbonise industries.

Most actors of the blue coalition argue that hydrogen is important or very important for decarbonisation. A large majority of actors including utilities and many industry actors advocate for a significant expansion of renewable energies in Germany whereas only very few actors think that hardly any renewable energy expansion is needed. This large support for renewable energy expansion in Germany differs strongly from earlier research results on renewables just a few years ago (Strunz et al., 2016). Interestingly, also actors from the coal and chemical industry, which have opposed renewable energies for long (Lauber & Jacobsson, 2016), now demand a rapid renewable energy expansion:

A hydrogen economy requires the accelerated and cost-efficient expansion of renewable energies and the accelerated expansion of electricity grids. (IG BCE & VCI, Mining, Chemistry and Energy Union & Chemical Industry Association).

Beliefs about the application of hydrogen in different sectors and the use of the natural gas grid differ and are important reasons for the observed heterogeneity within this coalition. The actors in the identified subset of the blue coalition express strong beliefs about using and expanding the existing natural gas grid for hydrogen while other actors did not position themselves on this topic. As the former come from the natural gas and / or the heating sector, their beliefs are in line with prior experiences.

With the BDEW it's clear how they position themselves, because they have the gas industry [as members], which above all wants to see hydrogen flowing through their grids. (EXP2, Research institute)

Actors within this subset of the blue coalition typically refer to “climate neutral gases” instead of hydrogen. These actors advocate for the use of blue or any hydrogen (including grey hydrogen) in all sectors and also for heating purposes. They advocate for blending hydrogen and natural gas while opponents would prefer to use the “expensive good” hydrogen in pure form.

Despite the early development stage of the subsystem, our interview partners already pointed to this emerging coalition (EXP2, EXP3, EXP4).

In the building sector, we see an unholy alliance between the building industry and the gas industry. The building industry has been resisting for years and decades that energy-efficient building renovation is promoted more strongly and, on the other hand, the gas industry is now afraid that it will lose its markets there. And heat is incredibly important for gas. (EXP4, NGO)

The other alliance is very, very clearly the fossil continuation together with the use in building heat. Building heat is the battlefield on which the future of the distribution grids will be decided. There is a very strong alliance here, which I have already felt. (EXP3, NGO).

Most conflicting policy core beliefs

Despite much consensus on developing hydrogen technologies, we identified three major lines of conflict. These include: (i) how to produce hydrogen, (ii) in which sectors and applications to use hydrogen and (iii) whether to formulate criteria for hydrogen import (strict criteria or few or no criteria, see blue, red and grey dots) (see Table 4).

Hydrogen production (x-axis, Fig. 5): Hydrogen can be produced from renewable energy sources (green hydrogen) or with natural gas and carbon capture technology (blue hydrogen). There is a clear divide between actors that favour green hydrogen (mostly e-NGOs, research institutes, energy associations, the automobile association, utilities, TSOs as well as Siemens Energy, MAN Energy Solutions and Thyssenkrupp) and those for whom blue hydrogen is also a viable option (most gas and industry actors). Some of the latter reject the “colour debate” about hydrogen production stating that any climate neutral form of hydrogen will be useful:

The current discussion about what colour hydrogen should have is not helpful: green hydrogen is produced by electrolysis with renewable energies, blue hydrogen is produced from natural gas, and turquoise hydrogen is produced by methane pyrolysis. All variants are climate-neutral because the CO₂ does not escape into the atmosphere. Since we will need large quantities of hydrogen in the future, the VDMA advocates the use of all forms of hydrogen that are climate-neutral. (VDMA, Mechanical Engineering Industry Association)

Next to these main groups, there are a few outliers. Three gas actors (Wintershall, Thüga and OGE) even accept hydrogen that is produced from fossil fuels without CCS.

Application of hydrogen (y-axis, Fig. 5): Many actors want to use hydrogen in a broad range of applications, including individual transport, and, partly, even heating. This group

of actors (lower part of the figure) includes industry actors, gas suppliers and three research institutes. Some of these actors also favour large hydrogen quantities.

I have the feeling that the Federal Ministry of Economics and Technology, the Ministry of Transport and many industry representatives are actually part of a very large coalition whose goal is simply to make the issue a big one. The goal could be: the greater the demand, the greater the export potential for technology from Germany and so on, the bigger the cake for everyone. (EXP1, Research institute)

The opposing view is typically expressed by e-NGOs, electric utilities, TSOs, Thyssenkrupp and MAN Energy Solutions: hydrogen should only or primarily be used as the last resort for decarbonisation in a limited range of applications. Some of these actors also make a connection between production and the application of hydrogen:

...caution is advised - hydrogen and PtX materials are not panaceas. Only if they are produced under strict sustainability standards and entirely from renewable energies can they serve as a complementary means of reducing emissions in certain applications. In many areas, however, their use would be inefficient and costly. An accelerated shift to renewable energies, the reduction of energy consumption, energy efficiency increases in all sectors and a comprehensive mobility transition remain the most important challenges in climate protection- and must be a clear priority for politics. (DUH)

Imports (colours, Fig. 5): A similar picture emerges if we look at the policy core beliefs regarding the import of hydrogen. Some actors argue that there should be strict standards for hydrogen imports (blue dots): it should be produced from renewable energy sources, and it should be beneficial for the exporting countries (e.g. in terms of social justice). These positions are put forward by e-NGOs, some research institutes and, not surprisingly, the association of renewable energy producers. A majority of actors, however, formulate less ambitious import conditions (red dots), or do not mention import standards (grey dots).

Summary

Our analysis revealed a high level of consensus on key belief dimensions, even though the policy subsystem and the associated policy issues are new, and the actors come from different sectoral backgrounds. In fact, most actors regard climate change as important or very important and support the net-zero target. Most also view the development of hydrogen technologies as a key strategy for decarbonisation. Some actors, mostly e-NGOs, take a more critical stance though as they highlight e.g. high costs and energy losses when producing hydrogen. It is interesting to see that the support for hydrogen reaches across all sectors, i.e. does not seem to be affected by past experiences. Next to this broad consensus, we identified three emerging conflicts: how hydrogen is produced, where it is used and whether to apply strict standards for imports or not.

We found two main coalitions: a larger one that includes many major industry associations and gas suppliers (including a gas-centred sub-coalition), and a smaller one with renewable energy associations, e-NGOs, electricity grid operators and two industry players. There is a high degree of heterogeneity within¹¹ and little distance between the coalitions. This is in line with earlier findings about coalitions in nascent subsystems not necessarily showing clear and expectable boundaries (Ingold et al., 2017; Nohrstedt et al.,

¹¹ Especially in the larger coalition.

2020). However, there are also unexpected findings such as the smaller coalition, in which e-NGOs and industry actors seem to form an ‘unusual alliance.’ We explore this further below.

Discussion

Here, we discuss our findings in the light of earlier studies. We draw on the key elements of our framework and we also discuss advantages and limitations of our approach.

Broad support for a novel technology: The ACF stipulates that policy core beliefs only change very slowly (Jenkins-Smith et al., 2017), which means that they may stand in the way when actors are confronted with novel and partly disruptive technologies. The transitions perspective arrives at a similar conclusion: Incumbent industry actors have often been observed to resist change (Smink et al., 2015; Wesseling et al., 2014) and to defend existing practices rather than embracing novel technologies (Hess, 2014; Lauber & Jacobsson, 2016; Stefes & Hager, 2020). So how to explain the extraordinary support for hydrogen?

Both technology and sector characteristics play a role here. In sectors such as steel or chemicals, incumbents have hardly any low-carbon technology alternatives to hydrogen. In the gas and heat sector, there are alternatives (heat pumps), but hydrogen is viewed as a way to also use existing assets (gas infrastructure) in the future (Ohlendorf et al., 2023). So, industry actors support hydrogen because it aligns well with their sectoral background and/or the limited technology alternatives within the sector. Technology characteristics also explain policy core beliefs such as those around a strong role for government: a new and expensive technology, especially with major infrastructural needs, cannot be realised without significant financial support by the state. These findings are in line with the acknowledgement of technology-related beliefs in earlier ACF research (Lindberg & Kammermann, 2021; Markard et al., 2016; Schmid et al., 2019).

Unusual ‘alliances’: Prior ACF research on environmental topics found that e-NGOs and think tanks typically form an ‘environmental’ coalition, while firms and industry associations are rather positioned in an ‘economic’ coalition (Haukkala, 2018; Leipprand et al., 2017; Markard et al., 2016; Hirschl, 2008). This speaks to the ACF idea of former ties being important. Our findings resemble the latter: both coalitions are made up of a mix of industry actors, associations and think tanks. This is what we call unusual alliances. It was particularly surprising to see the firms Thyssenkrupp (steel, machinery), and MAN Energy Solutions (trucks, machinery), 50Hertz and Amprion (TSOs) in a coalition together with environmental NGOs.

We can explain unusual alliances by technological characteristics and the sectoral backgrounds of actors. Hydrogen offers new business opportunities for large plant manufacturers such as MAN Energy Solutions and Thyssenkrupp, whose technological experiences and capabilities (and the associated beliefs) are well aligned with the new technology. Similarly, TSOs such as 50Hertz and Amprion but also the utilities RWE and Vattenfall are deeply embedded in the electricity sector (renewable energy production, grid operation). On top of that, they benefit from a growing demand for hydrogen (which is rather power intensive if produced by electricity). For e-NGOs, green hydrogen is the only form of hydrogen they support—with caution though. This is well aligned with their (former) policy core beliefs (e.g. around resource protection and sufficiency) in other policy fields.

Expectable alliances Earlier studies on emerging subsystems have shown that trust and former ties play a role in coalition formation and that actors seek to speak with one voice (Ingold et al., 2017; Nohrstedt et al., 2020). We can corroborate this through evidence on joint position papers (Table 2). These point to ‘expectable’ alliances of actors who have cooperated since long in the same sector such as the gas actors or the union IGBCE and the chemical industry association VCI.

While the ACF perspective would be sufficient to explain expectable alliances, our combined approach also highlights sectoral differences which might be needed for future developments: If each group of actors wants to use hydrogen in their specific sector and application, there might not be enough hydrogen for all, some actors suggest using it where it is needed most. So, we might see situations, in which existing ties break up because of competing interests.

Emerging conflicts The conflicts identified above do not run between actors that are in favour or against a technology or policy (Leipprand & Flachsland, 2018; Schmid et al., 2019), but unfold around how questions around production, application, and imports.

Beliefs on production and application are strongly influenced by sectoral and technological characteristics. The latter mostly depends on a preference for a broad application of hydrogen. This may indicate that how questions and finding a common solution reaching decarbonisation become more important than sticking to prior contacts.

Both, technological and sectoral characteristics seem to be particularly important in situations, in which former ties and existing contacts are of limited relevance for the underlying policy issue. Our empirical evidence for both, expected and unusual alliances, thus supports our combined conceptual endeavour. Our combined perspective elucidates belief and coalition formation in nascent subsystems in contexts where explaining change becomes more relevant than stability.

The emerging beliefs in a nascent subsystem may have broader effects. They may spill over to an adjacent policy subsystems and contribute to policy change (Jones & Jenkins-Smith, 2009; Meckling & Goedeking, 2023). A newly formed coalition in a nascent policy subsystem may influence policies in the system of origin, to which actors ‘return’ with their (new) beliefs (possibly presenting a new solution to an unresolved problem). The new ties that develop between actors in unusual coalitions may further lead to new collaborations advocating for novel solutions in adjacent policy subsystems, possibly even triggering (major) policy change.

Our approach advances theory by combining two strands of research that, independently from each other, started to analyse actor and policy dynamics in multi-system settings. We concentrated on a few key elements from each theoretical perspective, but we believe that incorporating other elements such as the ACF’s policy learning (Goyal & Howlett, 2020) might enhance our understanding of these processes, particularly in terms of their future development when analysing longer time frames.

Our study addresses the emergence of coalitions and a new subsystem at the intersection of several sectors. It may thus also inform policy integration analyses which address cross-sectoral policy problems (Kefeli et al., 2023; Milhorance et al., 2021). Policy integration is mainly understood as a process of multi-dimensional policy and institutional change (Candel & Biesbroek, 2016). Trein et al. (2023) advocate for a stronger focus on actor-oriented theories in policy integration studies to which the ACF would contribute (Kefeli et al., 2023). We think cross-sectoral analyses might also profit from paying more attention to technological change and transition dynamics. At the same time, the fine-grained coalition concept including the belief system represents a toolset that could inform transition studies

to engage in more detailed analyses of actors, actor groups and coalitions accelerating the transition and to identify their lines of conflict (Köhler et al., 2019).

This study is one of the first to address decarbonisation to net-zero from a multi-system, -technology and -actor perspective. However, hydrogen is but one example of a new subsystem bridging existing ones and we expect multi-system interactions (or ‘multi-sector policy subsystems’) to increase in the net-zero context (e.g., smart cities, low-carbon lifestyles, biodiversity or carbon capture and storage (Finstad & Andersen, 2023)). Innovations, new technologies and their characteristics will be important in most of these, as well as (new or changed) roles of incumbents and their interactions with challengers. While our observations point to the emergence of a new subsystem (see 3.2), this does not mean that the nascent subsystem will stay independent forever. If energy policies are restructured to better account for multi-system interactions or, in practical terms, sector coupling (Löhr & Chlebna, 2023), the hydrogen subsystem might as well be absorbed by the energy subsystem, e.g. when hydrogen strategies are no longer defined separately but as part of energy policies. This remains to be seen.

For policymakers, the multi-system setting comes not only with more diverse groups of actors but also with a wider range of beliefs and interests. New alliances then do not necessarily form along prior contacts or sectors of origin but along new policy- or technology-specific dimensions. Our study identifies lines of conflict which policymakers need to address to ensure a smooth development of hydrogen in Germany and elsewhere. At the same time, it points to the limits of a fast and broad application of green hydrogen.

The identified conflicts are also visible in the updated hydrogen strategy from 2023 (BMWK, 2023). The new strategy differs from the first, as some non-green hydrogen production technologies can now receive limited support during market formation. This means that the production method of hydrogen remains a point of conflict. The new strategy focuses on using hydrogen in hard-to-decarbonise sectors, including aviation and shipping, which conflicts with actors who also want to use it in heating. Political struggles around hydrogen production and use, referred to as ‘contested pathways’ by Kern et al. (2023), will continue.

We close this section with a brief reflection about the limitations of our study. While we have demonstrated the utility of a combined framework in one case, further studies will be needed on other cases. Also, due to our focus on the dynamics of a nascent subsystem, future research might want to explore how multi-system challenges play into more mature policy subsystems like water (Brandenberger et al., 2022). Another limitation is that we could not include all relevant actors into the formal parts of our analysis, as position papers were not always available (e.g., no position papers for the governing political parties or the federal government).¹² We chose to rely on a single data source to ensure consistency, comparability, and reproducibility. Future studies might consider using a variety of sources to include a broader range of actors, potentially revealing further sub-coalitions. Also, we have not gathered systematic data on coordination between actors. We have some insights through joint position papers and the expert interviews, but these remain limited. Even though coordination is only about to emerge in a nascent subsystem and thus harder to study, future research might want to explore coordination systematically, e.g. with a survey (Stritch, 2015). Finally, we carried out a study for one point in time (because hydrogen is

¹² For an analysis of the discourse around hydrogen in Germany which covers a greater number of actors please refer to Ohlendorf et al. (2023) and to Belova et al. (2023).

new on the agenda). Longitudinal studies can be promising once the subsystem is more developed, to possibly uncover changes in beliefs and coalitions.

Conclusion

Grand sustainability challenges often confront a broad range of different sectors, and they span traditional fields of policymaking. Where new technologies are developed to address these challenges, new policy subsystems might emerge, in which actors tackle the associated policy issues. For subsystems that emerge at the interface of different sectors, we can expect a high level of complexity, e.g., due to the large number of actors with different backgrounds and experiences.

To address this complexity, our framework places the emerging subsystem into a context of existing (but changing) socio-technical systems (or sectors). We call this a multi-system setting. We offer three central explanations: First, beliefs are shaped by technology characteristics, including the novel technology the subsystem addresses as well as the technologies in the adjacent sectors. Second, as many actors in the new subsystem originate from the surrounding sectors, their beliefs are shaped by their sectoral backgrounds. Third, beliefs are also shaped by prior contacts and alliances.

Our framework combines and contextualizes a ‘classic’ ACF analysis (identifying beliefs, coalitions and policy subsystems) with elements from transition studies. In situations where entire sectors are changing profoundly (e.g., as a response to grand sustainability challenges or other crises), we see new policy and technology developments, and new subsystems emerging, with a large number and variety of actors seeking to influence policymaking. In such dynamic settings, beliefs are shaped developments in the past (e.g., former contacts, experiences, sectoral backgrounds, characteristics of established technologies) but at the same time, they are challenged, and possibly reshaped, by developments unfolding in the present (e.g., novel technologies undermining existing businesses, shifting sectoral boundaries, socio-political crises).

For these kinds of new challenges, it can be fruitful to adapt and remodel existing frameworks, sometimes even from different disciplines. Here, we showed one possible way of combining insights from transition research and policy analysis. Future studies might want to follow or challenge our approach as well as explore new avenues to tackle the new complexities.

Appendix

See Fig. 6.

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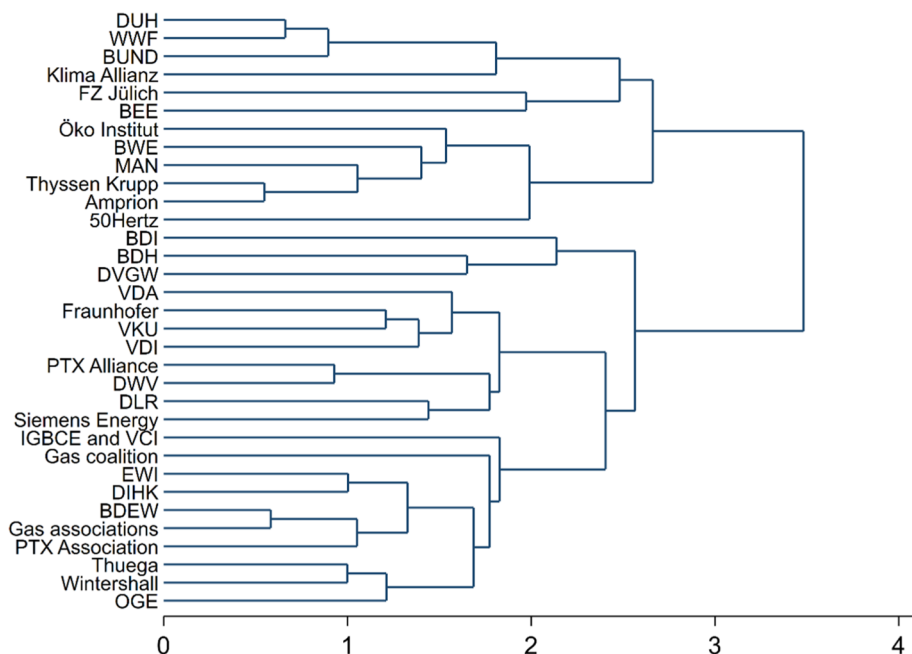


Fig. 6 Dendrogram of the cluster solution

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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